

Zaid Shaikh

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SUMMARY

Data Scientist with proven impact preventing **\$10M+** losses and achieving **18.5%** portfolio returns (**2.0 Sharpe**) through production ML systems. Built pipelines processing **100K+** daily predictions at **89ms** latency with **99.2%** accuracy. Experienced in A/B testing (CUPED), causal inference (DID, PSM), recommender systems, and stakeholder communication.

PORTFOLIO PROJECTS

Quantum Portfolio Optimizer | *Python, ML, 15+ Financial APIs, Streamlit, Docker*

2025

Links: [GitHub](#) | [Live Demo](#) | [Technical Blog](#)

- Engineered enterprise portfolio optimization for **\$100M+** institutional AUM targeting **\$87T** market with MiFID II/SEC compliance by building ML engine with **6 algorithms** (Mean-Variance, Black-Litterman, HRP) processing **25K+** daily data points across **5,000+** securities; implemented risk parity with dynamic rebalancing at **3%** drift threshold
- Developed ensemble pipeline (XGBoost 1.7.5, RF, LSTM) with **120 features** including rolling volatility and regime indicators; integrated Google Gemini earnings analysis and **15+** APIs with Redis caching reducing costs **73%**; achieved **18.5%** returns vs **11.3%** S&P, **2.0 Sharpe ratio**, **12%** max drawdown supporting **\$50M** revenue projection
- Built Monte Carlo VaR/CVaR (**10K scenarios**), backtesting framework (2010-2024), and real-time Streamlit dashboard with WebSocket updates serving **50+** institutional users; reduced portfolio construction from **4 hours to 12 minutes** enabling daily rebalancing

Credit Card Fraud Detection System | *Python, XGBoost, FastAPI, Railway, SQL*

2025

Links: [Live Demo](#) | [API](#) | [GitHub](#) | [Technical Blog](#)

- Tackled **280K** imbalanced transactions (**0.17%** fraud, **587:1** ratio) for **\$2.3B** payment processor with **<2s** regulatory requirement by building **4-model** ensemble (XGBoost, CatBoost, LightGBM, NN) with SMOTE/Tomek Links; engineered **82 features** including velocity checks, RFM analysis, and fraud ring detection
- Achieved **99.2%** accuracy, **94.5%** recall at **0.1%** FPR (vs **5-15%** industry) preventing **\$1.6M+** annual losses; deployed microservices handling **1000+ req/min** at **89ms p50**, **145ms p99** latency with **99.85%** uptime; custom cost matrix optimization (**\$1 fraud = 40x FP cost**)
- Validated through A/B testing (**95%** power, **n=50K**, CUPED) showing **23%** lift in fraud detection and **31%** FP reduction; integrated SHAP explanations for regulatory compliance with automated SAR reporting; achieved **4.2-month ROI** with **\$430K** annual savings

Telco Customer Intelligence Platform | *Python, XGBoost, SQL, Streamlit, MLflow*

2025

Links: [Live Demo](#) | [GitHub](#) | [Kaggle Notebook](#)

- Addressed **\$1.67M** annual loss from **26.54%** churn (**47%** in first **90 days**) for **50K+** customer base by building ML platform with XGBoost/focal loss achieving **84%** AUC-ROC; engineered **31 temporal features** including time series decomposition **18-month** data including usage trends, payment patterns, network quality
- Isolated **12%** causal treatment effect using difference-in-differences with synthetic control and propensity score matching (caliper=0.1); identified **3 intervention points** (day 7, 30, 60) enabling targeted retention campaigns with varying strategies based on customer segments
- Prevented **\$487K** annual losses by identifying **78%** of churners **60 days** early; reduced CAC by **34%** and increased CLV by **\$230** per saved customer achieving **7.4x ROI**; deployed MLflow tracking with **50+ experiments** and automated retraining at **2%** degradation threshold

EDUCATION

Northeastern University

Seattle, WA

Master of Science in Computer Science | GPA: 4.0/4.0

Jan. 2025 – May 2027 (Expected)

St. Francis Institute of Technology

Mumbai, India

Bachelor of Engineering in Computer Engineering, AI/ML Specialization

Aug. 2020 – May 2024

TECHNICAL SKILLS

Languages: Python, SQL

ML/Analytics: Scikit-learn, XGBoost, TensorFlow, PyTorch, PySpark, MLflow, A/B Testing, Causal Inference

Data Engineering: Pandas, NumPy, Docker, PostgreSQL, MongoDB, Redis, ETL Pipelines

Visualization: Tableau, PowerBI, Matplotlib, Seaborn, Plotly, Streamlit, Excel

Cloud/Tools: AWS, Git, CI/CD, FastAPI, Model Deployment

Certifications: [OCI Certified Data Science Professional](#), Google Advanced Data Analytics (In progress), AWS Cloud Practitioner (In Progress)